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Ekman Spirals with Variable Eddy Viscosity

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Contents

1 Introduction

The Ekman spiral is a famous consequence of the Coriolis force in meteorology and oceanography. Due to the interaction between shear forces in the atmosphere and ocean, the wind high up in the atmosphere, and wind over the ocean, will not give rise to a flow in the same direction at the surface. The adjustment of the flow from the surface will thus spiral, and is therefore properly named the Ekman spiral after its discoverer Ekman [1905]. The surface flow is at an angle from the flow aloft by more or less 45° [Ekman, 1905]. Due to this turning of the wind, the net flux of the wind will be perpendicular to the wind aloft. This is known as the Ekman transport and is crucial for understanding both meteorology and oceanography.

However, a surface wind-turning of 45° assumes a constant viscosity in the medium. This is unrealistic when a turbulent boundary layer is present. The boundary layer is around 1–2 km thick and is characterised by a presence of turbulence [Wallace and Hobbs, 2006]. The turbulence primarily arises from solar heating, causing instability and turnover, but also from mechanical turbulence from mean wind decreasing closer to the ground due to frictional shear forces. Although turbulence is non-linear, statistical analysis can provide useful first-order approximations, such as the downward flux being negatively and linearly proportional to the local gradient. Using this proportionality constant one can define the eddy viscosity, ν, for the boundary layer.

According to the mixing length hypothesis, the eddy viscosity should grow with the distance from the surface as the characteristic scales of the eddies grow. This can furthermore be described by a dimensional analysis of the boundary layer, where a logarithmic relation between the wind speed and height can be obtained [Wallace and Hobbs, 2006]. This showcases that the eddies, and thus the eddy viscosity, grow with height in the boundary layer. However, after a certain altitude, the turbulent boundary layer ceases to exist, and laminar flow takes over, which rapidly decreases the eddy viscosity. Thus, assuming that the eddy viscosity should increase initially from the surface and then vanish at a higher altitude is reasonable.

Following this logic, the viscosity should in reality vary with altitude. This would change the surface wind-turning from 45°, and thus also the Ekman transport. In this article, a varying viscosity with altitude will be studied under the Ekman equation. This will be done analytically and numerically, and will be compared to model data. This will give a deeper understanding of the Ekman spiral and provide insight when a surface angle different from 45° can be observed.

2 Theory

2.1 Problem formulation

The Ekman equation is derived from the steady-state Navier–Stokes equations on a rotating frame with shear flow. Assuming laplace friction, incompressible flow, negligible vertical motion, gives the horizontal momentum equations as

$$\begin{cases} -fv = -\dfrac{1}{\rho}\dfrac{\mathrm{d}p}{\mathrm{d}x} + \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}u}{\mathrm{d}z}\right), & \text{(1a)} \\ fu = -\dfrac{1}{\rho}\dfrac{\mathrm{d}p}{\mathrm{d}y} + \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu\dfrac{\mathrm{d}v}{\mathrm{d}z}\right), & \text{(1b)} \end{cases}$$

where u and v are the horizontal velocity components, f is the Coriolis parameter, ρ is the fluid density, and ν is the viscosity. Far away from the surface, shear flow and friction becomes negligible and the flow will approach geostrophic balance, $(u, v) \rightarrow (u_g, v_g)$, where the friction term in Equation 1b can be neglected. In this article, a Coriolis parameter of $1 \times 10^{-4}\,\mathrm{s}^{-1}$ and a zonal geostrophic wind will be used.

Subtracting the geostrophic balance from Equation 1b gives

$$\begin{cases} -f(v - v_g) = \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu \dfrac{\mathrm{d}u}{\mathrm{d}z}\right), & \text{(2a)} \\ f(u - u_g) = \dfrac{\mathrm{d}}{\mathrm{d}z}\left(\nu \dfrac{\mathrm{d}v}{\mathrm{d}z}\right). & \text{(2b)} \end{cases}$$

This problem can be formulated with one equation by introducing a complex velocity as

$$U \equiv u + iv, \tag{3}$$

and a corresponding complex geostrophic velocity. Using Equation 2b with the complex velocities U and U_g the coupled equations can be combined into the Ekman equation

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(\nu \frac{\mathrm{d}U}{\mathrm{d}z}\right) = if(U - U_g), \tag{4}$$

with the boundary conditions

$$U(0) = 0, \tag{5a}$$

$$U(z \to \infty) = U_g. \tag{5b}$$

The surface wind-turning angle can be determined by taking the argument of the infinitesimal change of the wind at the ground,

$$\theta = \arg\left(\left.\frac{\mathrm{d}U}{\mathrm{d}z}\right|_{z=0}\right). \tag{6}$$

The Ekman transport is defined as the mass flux per unit width for the entire Ekman spiral, which can be defined as

$$T = \int_0^\infty [U - U_g]\,\mathrm{d}z, \tag{7}$$

since the transport represents the net movement of the entire fluid column offset from the geostrophic wind aloft. The angle of the Ekman transport can consequently be determined by taking the argument, $\theta_T = \arg(T)$.

2.2 The vertical momentum flux

The Ekman equation can be reformulated by introducing a complex vertical momentum flux as

$$\tau = \nu \frac{\mathrm{d}U}{\mathrm{d}z}. \tag{8}$$

Using this definition, Equation 4 can be rewritten as

$$\boxed{\frac{\mathrm{d}^2\tau}{\mathrm{d}z^2} = i \underbrace{\lambda}_{\equiv \frac{f}{\nu}} \tau.} \tag{9}$$

Here, an eigenvalue λ has been introduced to the system. The new eigenvalue form is more intuitive since it directly compares the second order derivative with the variable itself. Furthermore, Equation 9 is similar to famous equations such as the Helmholtz equation and the time-independent Schrödinger equation, where λ becomes analogous to the potential. The new equation also has a clear physical interpretation, as it describes the vertical structure of the momentum flux in the Ekman problem. Using Equation 9, the boundary conditions become

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{10a}$$

$$\tau(z \to \infty) = 0. \tag{10b}$$

This new quantity will not change the surface angle, as the viscosity is always a positive real value and does not affect the argument,

$$\theta = \arg\left[\tau(z=0)\right]. \tag{11}$$

Solving for the Ekman transport becomes easier by using the momentum flux, as it directly reduces the integral of the Ekman transport. Observing the Ekman equation (Equation 4) it follows that $\frac{\mathrm{d}\tau}{\mathrm{d}z} = -if\left(U - U_g\right)$. Using the boundary condition that the momentum flux vanishes as infinity the Ekman transport reduces to

$$T = \int_0^\infty \left[U - U_g\right] \mathrm{d}z = \frac{i}{f}\tau(z=0), \tag{12}$$

which is proportional to the vertical momentum flux at the surface. Consequently, this can be interpreted as the surface stress at the ground being proportional to the Ekman transport of the fluid column.

2.3 The classical solution

The classical solution to the Ekman equation is obtained when the viscosity is assumed to be constant with altitude, $\nu =$ constant, and was found by Ekman [1905]. In this case the classical Ekman spiral solution is

$$\tau(z) = \nu U_g \frac{1+i}{h_{\mathrm{Ek}}} e^{-\frac{1+i}{h_{\mathrm{Ek}}}z}, \tag{13}$$

where

$$h_{\mathrm{Ek}} \equiv \sqrt{\frac{2\nu}{f}} \tag{14}$$

is the Ekman height scale. The surface angle of the classical Ekman theory gives 45°, since the solution is equally real and imaginary at $z = 0$. The transport yields an angle of 90° from the surface wind-turing, meaning that the transport and surface wind is perpendicular.

2.4 The two layer model

The next natural step from a constant viscosity profile is to assume two layers of constant viscosity in each layer. The lower level is labelled as having the viscosity ν_1 up until a height of H, where the next viscosity of ν_2 takes over. From this assumption a superimposed solution similar to Equation 13 is obtained as

$$\begin{cases} \tau_1(z) = A_1 e^{\frac{1+i}{h_{\mathrm{Ek1}}}z} + B_1 e^{-\frac{1+i}{h_{\mathrm{Ek1}}}z}, & \text{(15a)} \\ \tau_2(z) = B_2 e^{-\frac{1+i}{h_{\mathrm{Ek2}}}z} & \text{(15b)} \end{cases}$$

where the Ekman height scales are defined for each layer like in Equation 14. To solve for the prefactors above one has to use that the velocity and momentum flux is continuous, which gives the jump conditions

$$U_1 = U_2 \quad \text{at } z = H, \tag{16a}$$

$$\tau_1 = \tau_2 \quad \text{at } z = H. \tag{16b}$$

Using the jump conditions together with the boundary conditions in Equation 5 an expressions for the prefactors is obtained as

$$A_1 = -\nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{1-\gamma}{1-\gamma+e^{2(1+i)\varphi}\left(1+\gamma\right)}, \tag{17a}$$

$$B_1 = \nu_1 U_g \frac{1+i}{h_{\mathrm{Ek1}}} \cdot \frac{\left(1+\gamma\right)e^{2(1+i)\varphi}}{1-\gamma+e^{2(1+i)\varphi}\left(1+\gamma\right)}, \tag{17b}$$

$$B_2 = \nu_2 U_g \frac{1+i}{h_{\mathrm{Ek2}}} \cdot \frac{2e^{(1+i)\varphi\left(1+\gamma^{-1}\right)}}{1-\gamma+e^{2(1+i)\varphi}\left(1+\gamma\right)}, \tag{17c}$$

$$\text{where } \gamma = \sqrt{\tfrac{\nu_2}{\nu_1}} \tag{17d}$$

and where φ is the non-dimensional layer thickness defined as

$$\varphi = \frac{H}{h_{\mathrm{Ek1}}}. \tag{18}$$

This provides a complete expression for the vertical momentum flux in the two-layer model, as shown in Figure 1. The figure illustrates how the vertical momentum flux varies with altitude for different viscosity ratios and non-dimensional heights, φ. For an increasing viscosity with height, a larger surface stress in the zonal direction is observed. Conversely, for a smaller upper-layer viscosity, a larger meridional surface stress is evident. Observing the interface, H, reveals that for larger φ, the model behaves more like a single-layer model, as the second layer begins at a height where the Ekman spiral has already decayed. Interestingly, smaller values of φ yield a larger difference between the zonal and meridional wind stress.

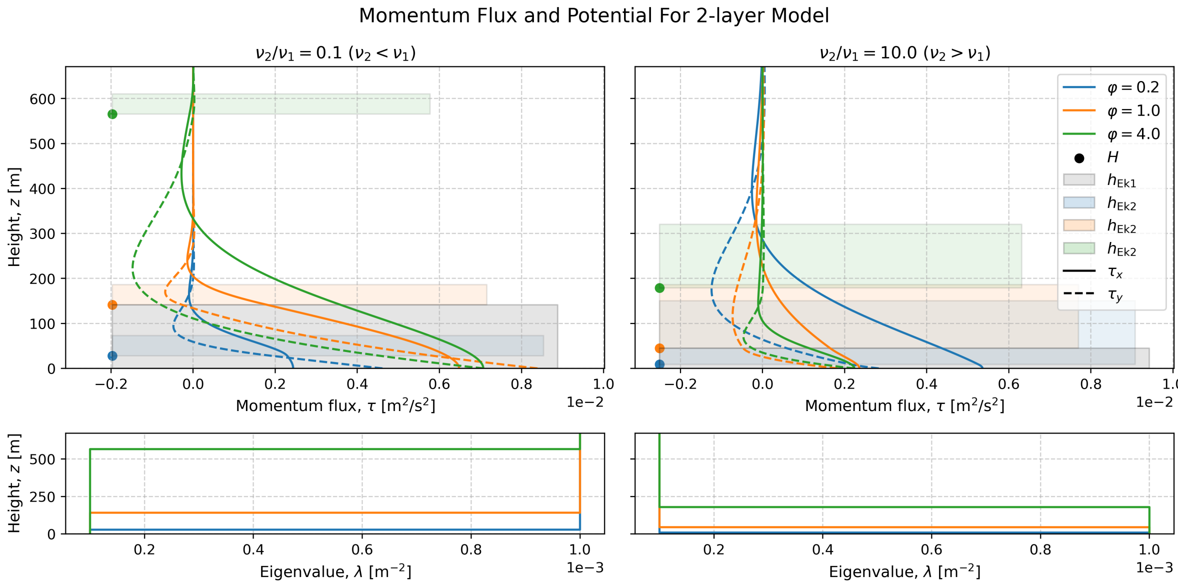

Figure 1: The vertical structure of the momentum flux and the eigenvalues for the two layer model. The momentum flux is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek1}}$ and the ratio of the viscosities. The corresponding Ekman heights and interfaces are shown with the vertical momentum flux.

By defining the constant

$$R = \frac{1-\gamma}{1+\gamma}. \tag{19}$$

and using the derived solution for the vertical momentum flux, the surface wind-turning angle becomes

$$\theta = \arg\left[(A_1 + B_1)\right] \tag{20}$$

$$= 45^\circ + \arg\left[\frac{e^{(1+i)\varphi} - Re^{-(1+i)\varphi}}{e^{(1+i)\varphi} + Re^{-(1+i)\varphi}}\right], \tag{21}$$

which can be observed as a function of φ in Figure 2. The solutions for different viscosity ratios can be observed, half of which have a higher upper-level viscosity and the other half have a lower upper-level viscosity. One can observe a convergence towards the classical Ekman value of 45° when $\varphi \to \infty$ for all the viscosity ratios. This can be attributed to the high altitude interface H not acting in the domain of the spiral, which reduces the problem to a one layer model.

The zero limit is, however, dependent on the viscosity ratio, where finite ratios approach 45°. This occurs when the interface H is close to the surface, again reducing the problem to a one layer model if $H = 0$. However, when $\frac{\nu_2}{\nu_1} \to 0$ and $\varphi \to 0$ the angle goes to $\theta \to 90^\circ$. Respectively, when $\frac{\nu_2}{\nu_1} \to \infty$ and $\varphi \to 0$, $\theta \to 0^\circ$. This indicates that the interface approaches the ground while the viscosity contrast becomes extreme, leading to a surface-dominated response where the surface stress aligns either perpendicular or parallel to the upper geostrophic flow.

In the case of weaker upper viscosity the perpendicular angle can be interpreted as the entire turning, due to the Coriolis force, has to occur at the surface since the viscosity becomes negligible aloft. In the opposite case, a weaker viscosity at the ground makes the Ekman spiral less prominent. Since the Ekman spiral occurs due to the surface stress if no major viscosity is present the stress becomes negligible and the wind-turning does not occur. For $\nu_1 > \nu_2$ (and $\nu_2 > \nu_1$ respectively) the solution will always provide a result greater than 45°, except between the $\varphi \in \left(\frac{\pi}{2}, \pi\right)$ were a angle below 45° can be observed.

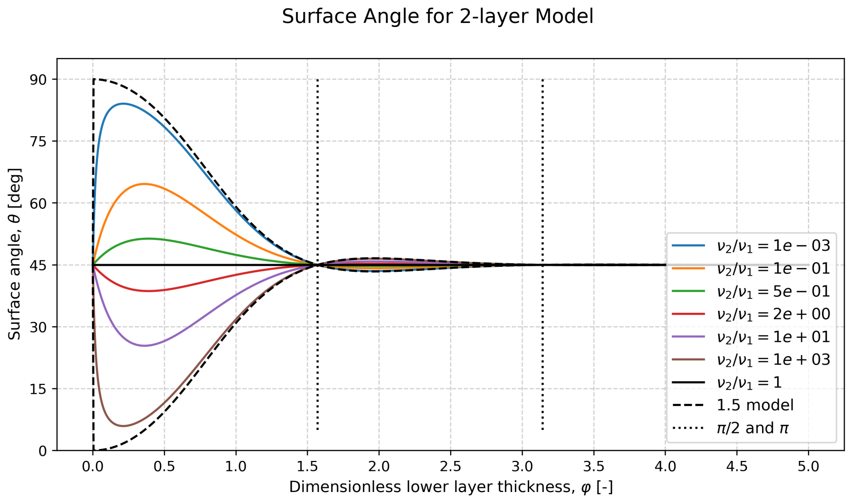

Figure 2: The surface angle as a function of the dimensionless layer thickness $\varphi = H/h_{\mathrm{Ek1}}$ for the two-layer model. Different coloured curves correspond to different layer viscosity ratios. The classical Ekman angle of 45° is shown as a reference.

The full vertical spiral of the two-layer model can be obtained by calculating the angle at each height in Figure 3. This spiralling compared to the classical Ekman theory can be observed in Figure 3. The angles deviate throughout the structure from the classical solution, but for the larger φ-value, a closer likeness to the classical solution can be observed. Interestingly, if the viscosity increases with altitude, the angles are lower than the classical solution at the surface, but increase to be larger than the classical solution higher up. The opposite is true for the other case, when the viscosity decreases with altitude.

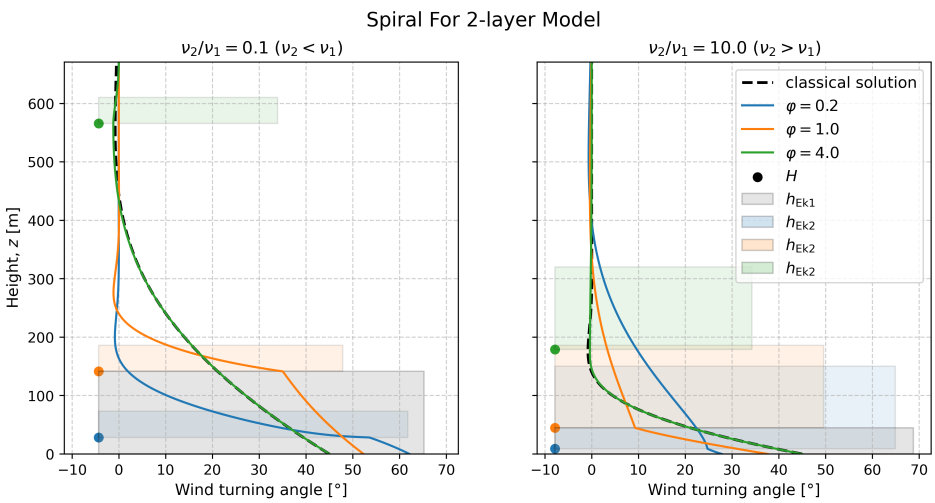

Figure 3: The vertical spiral of the velocities for the two layer model. The angle is dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek1}}$. The classical Ekman solution (when $\nu_{\mathrm{const}} = \nu_1$) is shown as a reference.

2.5 The exponential decaying viscosity model

Another viscosity profile that can provide a fully analytical solution is an exponentially decaying viscosity with altitude. Here the viscosity is set to decay from an initial viscosity of ν_0, with a decay rate of k as

$\nu(z) = \nu_0 \exp(-kz)$. Solving Equation 9 with this viscosity profile can be done by introducing the new variable

$$t = 2\,(1+i)\,\varphi_{\text{exp}}\, e^{\frac{kz}{2}}. \tag{22}$$

The non-dimensional parameter φ_{exp} resembles the non-dimensional layer thickness from the two-layer model, but defined as

$$\varphi_{\text{exp}} = \frac{1}{k\,h_{\text{Ek0}}} = \frac{1}{k}\sqrt{\frac{f}{2\nu_0}}. \tag{23}$$

Using these newly defined variables Equation 9 transforms into

$$t^2\frac{\mathrm{d}^2\tau}{\mathrm{d}t^2} + t\frac{\mathrm{d}\tau}{\mathrm{d}t} - t^2\tau = 0, \tag{24}$$

which is a differential equation with a standard solution. The standard solution is a modified Bessel equation as

$$\tau(t) = C_1 I_0(t) + C_2 K_0(t), \tag{25}$$

where C_1 and C_2 are constants, while $I_0(t)$ and $K_0(t)$ are modified Bessel functions of the first and second kind of order zero, respectively. By applying the previously defined boundary conditions from Equation 10 the two constants can be found. Applying the geostrophic boundary condition from Equation 10b shows that $C_1 = 0$ since $I_0(t)$ does not go to zero when $t \to \infty$ (which occurs when $z \to \infty$). Applying the surface boundary condition from Equation 10a an expression for the second constant can be found. Transforming the solution to depend on the variable z gives

$$\tau(z) = \frac{(1+i)\,fU_g}{2k\varphi}\frac{K_0\left(2\,[1+i]\,\varphi_{\text{exp}}e^{kz/2}\right)}{K_1\left(2\,[1+i]\,\varphi_{\text{exp}}\right)}. \tag{26}$$

This solution for the vertical momentum flux in the exponential model can be observed in Figure 1. There, the change of the vertical momentum flux with altitude for different non-dimensional heights, φ, can be observed. For smaller φ, the solution is more localized toward the surface, whereas larger φ produces a more extended spiral. This implies that for larger φ, the viscosity changes very little, causing the spiral to behave as in the one-layer model. In contrast, smaller φ indicates a rapid viscosity decrease from the ground.

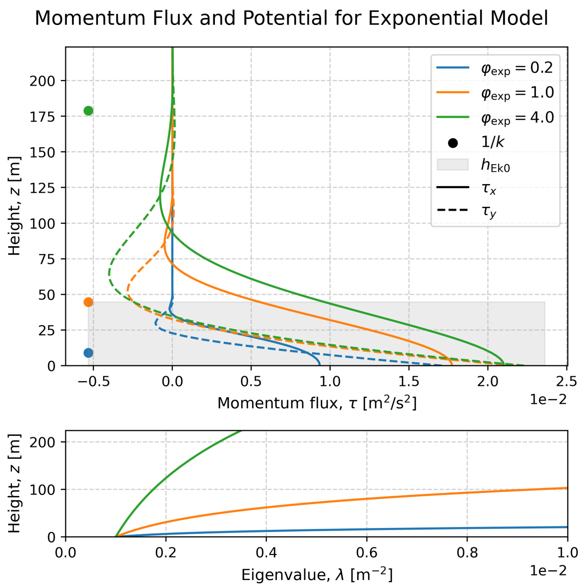

Figure 4: The vertical structure of the momentum flux for exponentially decreasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi_{\text{exp}} = 1/k\,h_{\text{Ek0}}$. The corresponding Ekman heights and eigenvalue are shown with the vertical momentum flux. The solution assumed a purely zonal geostrophic flow and a Coriolis parameter of $f = 1 \times 10^{-4}\,\text{s}^{-1}$

Using Equation 26 the surface angle of this model gives

$$\theta = 45^\circ + \arg\left[\frac{K_0\left(4\,[1+i]\;\varphi_{\text{exp}}\,\right)}{K_1\left(4\,[1+i]\;\varphi_{\text{exp}}\,\right)}\right], \tag{27}$$

where K_1 is the modified Bessel function of the second kind of order one. Figure 5 showcases the result from the surface wind-turning angle for this model. As this model only provides scenarios where the viscosity decreases from the ground, all angels are greater then the reference angle of 45° (following the logic from the two layer model). One can observe a similar limit as in Figure 2 when $\varphi \to 0$ for $\nu_2 = 0$, as Equation 27 also approaches 90°. The same limit is also true for $\varphi \to \infty$, which gives $\theta \to 45°$, although this exponential solution converges slightly slower than the two-layer solution. The reasoning behind this convergence is that the lower decay rates makes the solution behave more like the classical Ekman spiral with constant viscosity. The opposite case when the layer is really thin forces a strong turning close to the surface.

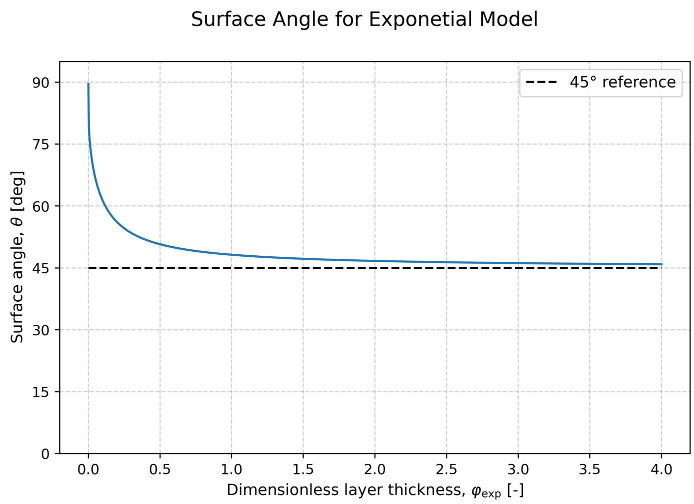

Figure 5: Surface angle for the exponentially decaying viscosity model as a function of the dimensionless parameter $\varphi = H/h_{\text{Ek}}$. The solution assumed a purely zonal geostrophic flow.

The full vertical spiral of the exponential model can, like in the two-layer model, be obtained at each height from Figure 4. The spiralling of the exponential model compared to the classical Ekman theory can be observed in Figure 6. The angles are higher than the classical solution at the surface but decrease to be smaller than the classical solution higher up, following the logic presented in the two-layer model.

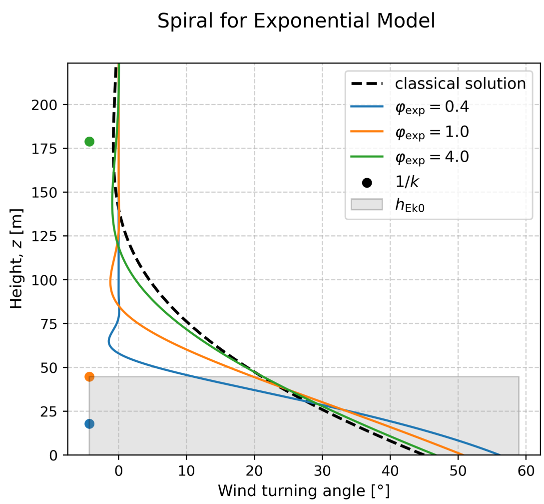

Figure 6: The vertical spiral for the exponentially decreasing viscosity model. The angle is dependent on the dimensionless parameter $\varphi_{\text{exp}} = 1/k\, h_{\text{Ek0}}$. The classical Ekman solution (for ν_0) is shown as a reference.

2.6 The simplified upper boundary condition model

In the previous sections two completely analytical solutions for two different altitude-varying viscosity profiles were derived. However, difficulty solving Equation 9 comes from the altitude dependence of the viscosity profile $\nu(z)$. This introduces variable coefficients into the complex second-order ordinary differential equation, which in general prevents closed-form analytical solutions. Analytical solutions therefore exist only for specific viscosity profiles that allow the equation to be transformed into a solvable form.

Equation 9 is furthermore also difficult to solve in a numerical setting, as the upper boundary condition is a limit. This will put limits on the viscosity profiles to study, as the viscosity cannot increase with altitude continuously. In the physical world, this makes sense, since we assume that high up in the atmosphere (or deep down in the ocean), the flow becomes laminar and the turbulent viscosity small. I.e. that the turbulent boundary layer has a limit, which is the same as the geostrophic limit. However, by limiting ourselves to only decreasing viscosity profiles, the real-world differential equation becomes impossible to solve.

A solution to solve this problem is to introduce a simplified upper boundary condition. For the two-layer model, this is reduced to a 1.5 layer model, where we assume that the upper-level viscosity is negligible ($\nu_2 \approx 0$), which could be observed as the limit in Figure 2. This also makes physical sense, since we know that both the atmosphere and the ocean consist of a boundary layer where turbulent mixing takes place. Allowing the viscosity to vary freely in the lower layer reduces the problem to a simplified upper boundary condition model. This will allow viscosity profiles that do not converge at $z \to \infty$, since the new domain is between $z \in [0, H]$. This modifies the boundary conditions in Equation 10 as

$$\left.\frac{\mathrm{d}\tau}{\mathrm{d}z}\right|_{z=0} = -ifU_g, \tag{28a}$$

$$\tau(z = H) = 0. \tag{28b}$$

Above, H becomes the height of the boundary layer, and the upper boundary condition was argued from the jump condition in the two-layer model, as seen in Equation 16b.

2.6.1 Simple viscosity profiles for the SUBC model

Using the the simplified upper boundary condition, SUBC model, simple viscosity profiles can be studied and evaluated numerically. The easiest, except for the constant viscosity, is a linearly varying viscosity with height as

$$\nu_\uparrow(z) = \frac{\nu_{\max}}{H} z, \tag{29a}$$

$$\nu_\downarrow(z) = \frac{\nu_{\max}}{H}(H - z), \tag{29b}$$

where $\nu_{\max}$ is the maximal viscosity. For both of these profiles a dimensionless thickness coordinate can be defined as

$$\tilde{z} = \frac{z}{H} \implies \begin{cases} \nu_\uparrow(z) = \nu_{\max}\tilde{z} \\ \nu_\downarrow(z) = \nu_{\max}(1 - \tilde{z}). \end{cases} \tag{30}$$

This new coordinate changes the previous z-derivative, by introducing a factor of $1/H$ when taking the derivative with respect to $\tilde{z}$. Using these linear viscosity profiles, together with the new dimensionless thickness coordinate, Equation 9 is rewritten for the two profiles as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{\tilde{z}}}_{\equiv 1/\tilde{\nu}_\uparrow} \underbrace{\frac{H^2 f}{\nu_{\max}}}_{=2\varphi^2} \tau, \tag{31a}$$

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i \underbrace{\frac{1}{1-\tilde{z}}}_{\equiv 1/\tilde{\nu}_\downarrow} \underbrace{\frac{H^2 f}{\nu_{\max}}}_{=2\varphi^2} \tau. \tag{31b}$$

Above the dimensionless thickness φ was again introduced following the logic from before.

Another simple profile to evaluate is a parabolic increasing viscosity as

$$\nu_p = \frac{4\nu_{\max}}{H^2}(H - z)\, z. \tag{32}$$

Again, using the non-dimensional height coordinate $\tilde{z}$ and the dimensionless thickness φ, Equation 9 is rewritten as

$$\frac{\mathrm{d}^2\tau}{\mathrm{d}\tilde{z}^2} = i\,\underbrace{\frac{1}{4\tilde{z}\,[1-\tilde{z}]}}_{\equiv 1/\tilde{\nu}_p}\underbrace{\frac{H^2 f}{\nu_{\max}}}_{=2\varphi^2}(U - U_g). \tag{33}$$

Since the height coordinate is normalized for the height H, the dimensionless thickness simply becomes the same as the inverse of the normalized Ekman scale height as

$$\varphi = \frac{1}{\tilde{h}_{\mathrm{Ek}}}. \tag{34}$$

3 Methodology

3.1 Numerical method

For arbitrary viscosity profiles, Equation 9 was shown to not admit a closed-form analytical solution. The equation was therefore solved numerically as a boundary value problem using the `solve_bvp` routine available in `SciPy` [Virtanen et al., 2020].

For the SUBC model, the equation was shown to be expressed in terms of a dimensionless layer thickness by using an height-normalized eddy viscosity profile, $\tilde{\nu}(z)$. Separating Equation 31a, 31b and 33 into zonal and meridional components yields

$$\frac{\mathrm{d}^2\tau_{\mathrm{x}}}{\mathrm{d}z^2} = -\frac{1}{\tilde{\nu}(z)}2\varphi^2\tau_y, \tag{35a}$$

$$\frac{\mathrm{d}^2\tau_{\mathrm{y}}}{\mathrm{d}z^2} = \frac{1}{\tilde{\nu}(z)}2\varphi^2\tau_x. \tag{35b}$$

To apply the numerical boundary value solver, the equations were rewritten as a first-order system by introducing the new variables

$$\tau_x' = \frac{\mathrm{d}\tau_x}{\mathrm{d}z}, \qquad \tau_y' = \frac{\mathrm{d}\tau_y}{\mathrm{d}z}. \tag{36}$$

Expanding Equation 35b into four coupled first-order equations gives

$$\begin{cases} \frac{\mathrm{d}\tau_x}{\mathrm{d}z} &= \tau_x' \\ \frac{\mathrm{d}\tau_x'}{\mathrm{d}z} &= -\frac{2\varphi^2\tau_y}{\tilde{\nu}(z)} \\ \frac{\mathrm{d}\tau_y}{\mathrm{d}z} &= \tau_y' \\ \frac{\mathrm{d}\tau_y'}{\mathrm{d}z} &= \frac{2\varphi^2\tau_x}{\tilde{\nu}(z)}, \end{cases} \tag{37}$$

with the boundary conditions

$$\tau_x'(\varphi = 0) = 0, \qquad \tau_x(\varphi = 1) = 0, \tag{38a}$$

$$\tau_y'(\varphi = 0) = -fu_g, \qquad \tau_y(\varphi = 1) = 0. \tag{38b}$$

If the viscosity profile, $\tilde{\nu}$, includes a zero value in the domain, then the coupled Equation 37 becomes unsolvable since we divide by zero. In this scenario one has to introduce a constant background viscosity, ϵ as

$$\hat{\nu} \to \hat{\nu} + \epsilon, \quad \text{where } \epsilon = \frac{\nu_{\mathrm{background}}}{\nu_{\max}} \tag{39}$$

and $\hat{\nu}/\epsilon \gg 1$ is assumed. This background viscosity was thus applied for the linear decreasing, linear increasing, and the parabolic model since they have zero viscosity at the surface (or at the top).

The SUBC model equations were solved iteratively on a vertical grid. The solver required an initial guess, which was chosen based on the initial setup of the system. The initial guess was therefore set as the classical solution for a constant viscosity. The final solution was found when the iterative solution reached a steady state for the differential equations with the imposed boundary conditions.

4 Result

4.1 The vertical structure of the momentum flux

The vertical profiles from the numerical SUBC model can be viewed in Figure 7–9 together with the normalized Ekman height for the different layers. An important note is that when $\varphi < 1$ then the Ekman heights become larger than the limit H. This is the same as stating that the entire spiral occurs really close to the surface.

Since the surface wind-turning angle is calculated by comparing the initial change of the viscosity, both the linear increasing and the parabolic models are dependent on the background viscosity ϵ. The effect of varying ϵ can be observed in Figure 8–9. However for the linear decreasing model, no major effect could be observed by varying the background viscosity. Thus, a background viscosity of 1×10^{-6} was used for this specific model.

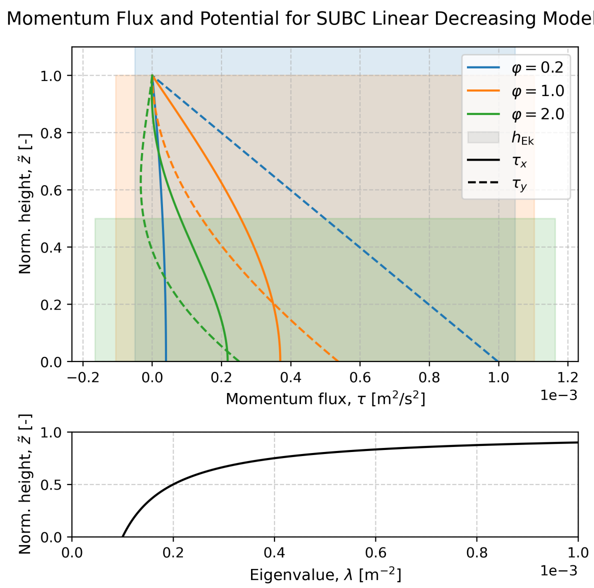

Figure 7: Numerically computed vertical structure for linearly decreasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = \tilde{h}_{\mathrm{Ek}}^{-1}$. The corresponding Ekman heights and eigenvalue are shown with i the figure.

4.2 The surface angles

Figure 10–12 show the surface angles from the numerical SUBC models. Similarities between these models, the decreasing two-layer model, and the exponentially decaying model can be observed. They all have a lower limit of 90° and an upper limit of 45°. The reason for the similarly between the exponential model and the increasing two layer model, is that the SUBC model is based on the decreasing two layer model, as $\nu = 0$ above the height H.

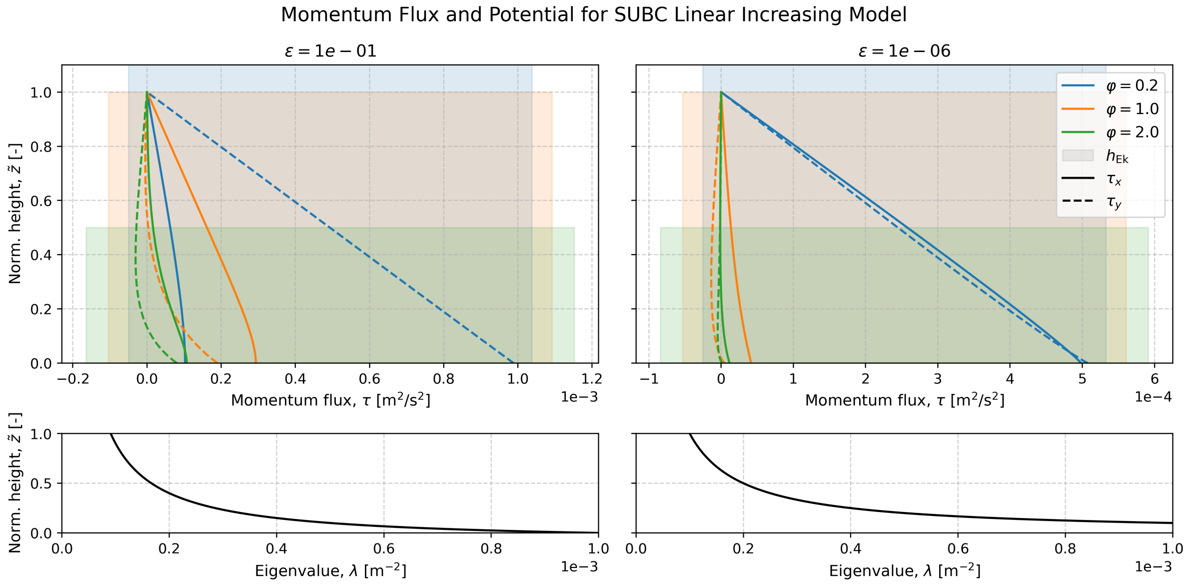

Figure 8: Numerically computed vertical structure for linearly increasing viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ. The corresponding Ekman heights and eigenvalue are shown with i the figure.

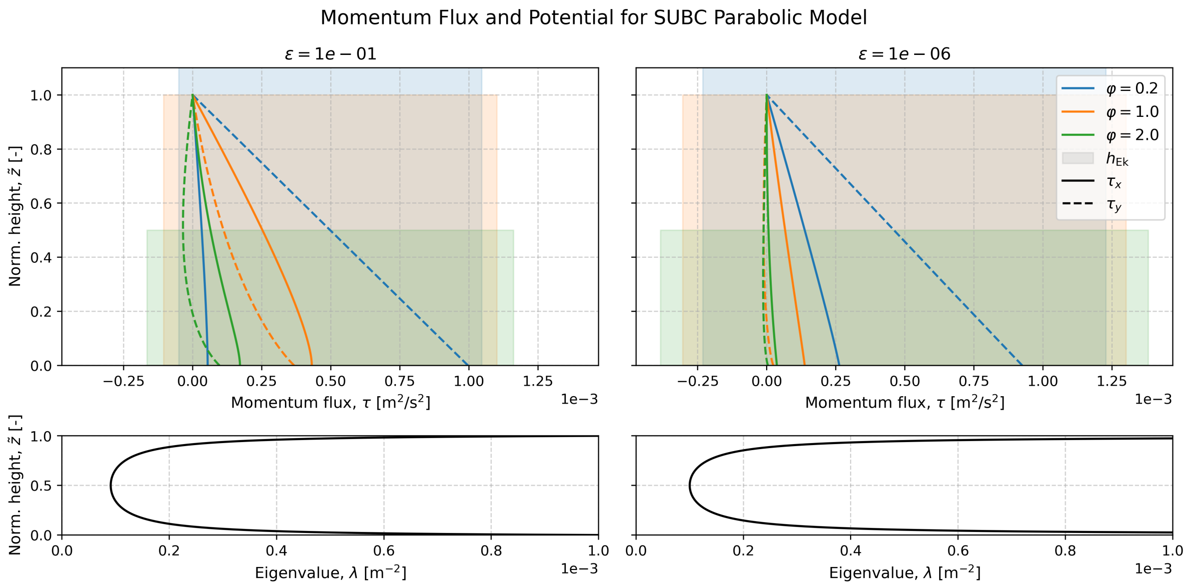

Figure 9: Numerically computed vertical structure for parabolic viscosity. The momentum flux is dependent on the dimensionless parameter $\varphi = \tilde{h}_{\mathrm{Ek}}^{-1}$ and the background viscosity ϵ. The corresponding Ekman heights and eigenvalue are shown with i the figure.

Thus, in the limit when $\varphi \to 0$ the Ekman height becomes infinitely larger then the interface H, providing a 1.5 layer like solution which has the the limit of 90°. When $\varphi \to \infty$ the maximum value of the profile decreases, providing a smaller change with altitude, and a more constant like viscosity. Thus, the solution approaches the classical Ekman solution, with an angle of 45°.

The result from the linearly decreasing SUBC model is shown in Figure 10, where the values smoothly transition from $90° \to 45°$ as φ increases. This plot resembles the exponential decaying model seen in Figure 5 and the 1.5 model as seen in Figure 2. Again, all angles are larger then the reference angle of 45°. The close resemblance to the exponential and 1.5 layer solutions makes sense since this solution is also a strictly decreasing solution.

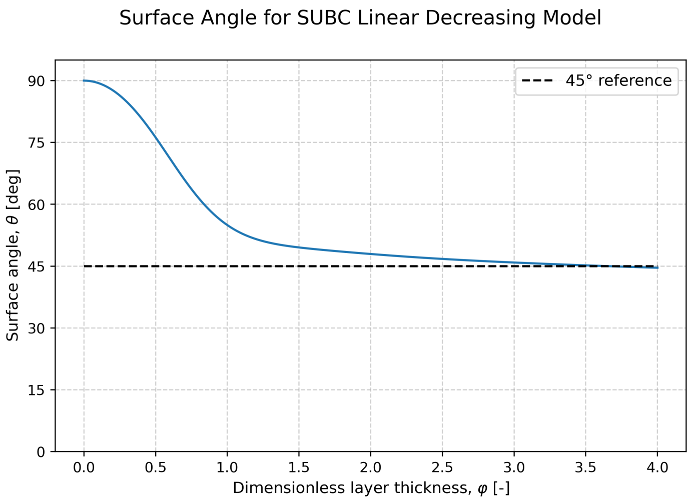

Figure 10: Numerically computed surface angles for decreasing viscosity profiles in the SUBC model. The angles are dependent on the dimensionless parameter $\varphi = H/h_{\text{Ek}}$.

For the linearly increasing viscosity in Figure 11 a different result can be seen. The upper limit is again located at 45°, but at a very large non-dimensional height. For smaller background viscosities the limit occurs at many magnitudes higher then for the previous models. Since the eigenvalue is proportional to the inverse o the viscosity, smaller background viscosities yields much larger eigenvalues (as seen in Figure 8). Thus, this limit of 45° occurs at much higher non-dimensional heights.

The curve obtains values smaller then 45° outside of the limits discussed above. Since the limits are the extreme cases, this interior is the more interesting part of the result. The surface wind-turning angle is smaller then 45° since a larger viscosity at a higher altitude means that the turning of the wind is smaller near the surface and greater aloft. This forces the surface angles to become less then 45° near the ground.

Another interesting observation is that, away from the limits, she surface wind-turning angle is more dependent on the background viscosity rather then the non-dimensional height, especially for the lower background viscosities. This occurs since the eigenvalue is the inverse of the viscosity and that the initial increase from the ground is the most crucial for the surface angle. Thus, if the background viscosity is much smaller then the maximal viscosity then a smaller ϵ is obtained and the initial eigenvalue jump from the surface becomes greater. In the theoretical limit when $\epsilon \to 0$, the initial eigenvalue jump would be infinite, and the surface angle would be zero.

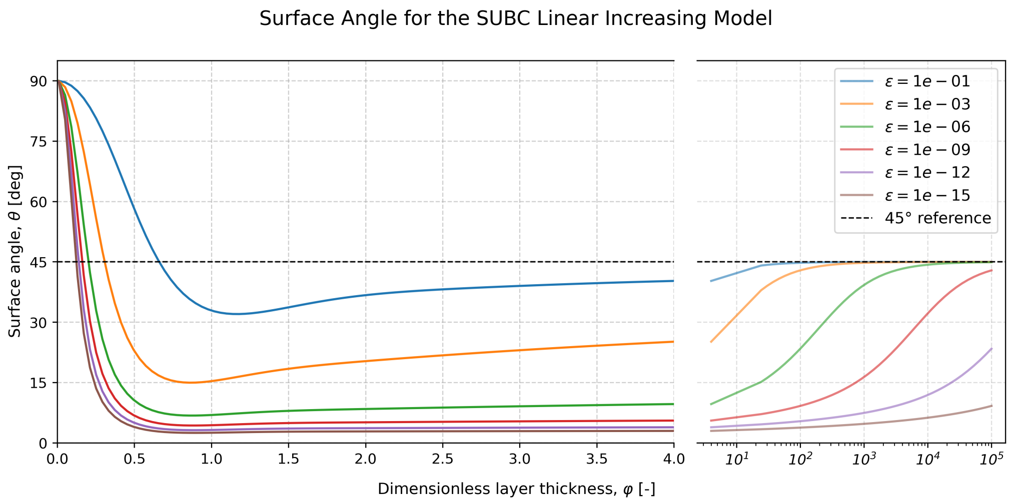

Figure 11: Numerically computed surface angles for linearly increasing. The angles are dependent on the dimensionless parameter $\varphi = H/h_{\text{Ek}}$ and the background viscosity ϵ.

The SUBC model with a parabolic viscosity profile gave Figure 12. Very similar behaviour to the increasing viscosity profile in Figure 11 can be observed. Since the viscosity jump from the surface is what determines the surface angle, the parabolic decrees aloft has no dramatic effect on the surface angle. The two models differ slightly in that the parabolic curves cross the 45° later than the linearly increasing curves.

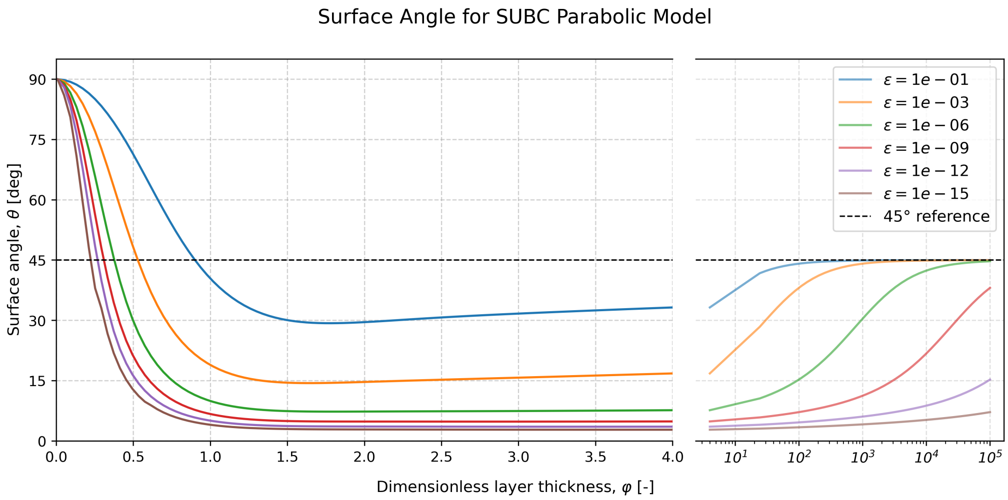

Figure 12: Numerically computed surface angles for a parabolic viscosity profile in the SUBC model. The angles are dependent on the dimensionless parameter $\varphi = H/h_{\mathrm{Ek}}$ and the background viscosity ϵ.

4.2.1 The Ekman transport

Computing the transport with the surface vertical momentum flux as Equation 12 yielded exactly the same shapes as seen in Figure 2–12, but centred around 135° rather then 45°. The angle differences between the surface angles and the transport angle was thus always 90°, just like in the classical Ekman theory.

5 Discussion

Observing the analytical and numerical models, angles between 0° to 90° could be observed. Generally, surface wind-turning angles above the classical value of 45° could be observed with strictly decreasing viscosities with altitude. The only case where this did not occur is for a small region in the two-layer model. However, since this region only had a lowest value of 43.4°, the importance of this small region can be dismissed. The reasoning behind decreasing viscosities with altitude providing surface angles over the classical value can be attributed to the decreasing viscosity forcing the rotation due to the Coriolis force to occur at a lower latitude. As the viscosity is initially large but decreases afterwards, the spiral is forced to turn more at the lower altitudes. This has to occur in order to satisfy the geostrophic limit above. Thus, the surface angle becomes larger than the classical value for the decreasing viscosities with altitude.

For the increasing viscosities with altitude, the opposite could be observed with smaller surface wind-turning angles than the classical value. For the two-layer model, only a small region had angles greater than the classical value. However, the largest value was only measured at 46.6°, which is close to the classical limit. The same logic follows to explain why this surface angle became smaller, as the explanation for why the surface angles became larger for the decreasing viscosities with height. If the viscosity increases with height, then the spiral has to turn slower near the surface in order to turn enough to reach the geostrophic limit aloft. And since the viscosity increases with altitude, more turning can occur there, as it is the shear due to the viscosity which forces this turning.

The same reasoning can be used to explain the limits for the surface angles when the dimensionless layer thickness goes to infinity or zero. Since the dimensionless layer thickness is the ratio between the characteristic height and the Ekman height scale, it follows that when the ratio goes towards infinity, a constant viscosity is obtained, and the angles go to the classical value of 45°. Furthermore, when the ratio between the characteristic height and the Ekman height scale goes towards zero, one would obtain the extremes of the logic presented in the paragraphs above. Since the Coriolis force is perpendicular to the flow, a 90° surface angle will be obtained if the viscosity decreases with altitude, since the entire rotation has to occur at the surface. On the other hand, if the viscosity increases with height, then no part of the rotation can occur at the surface, and a parallel angle of 0° will be obtained at the surface.

6 Conclusion

A The Source Code

The source code for this article can be found on GitHub under:

`https://github.com/FredrikBergelv/Ekman-Spiral-Project`

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